Probabilistic Modelling of Motor Subtype Progression in Parkinson's Disease

Project Report

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Abstract

Parkinson's disease patients can be classified into Tremor Dominant (TD), Indeterminate and Postural Instability / Gait Difficulty (PIGD) motor subtypes using scores derived from the MDS-UPDRS examination. This classification is useful for describing a patient's current motor characteristics, but it does not describe how the subtype may change over time.

This project investigates whether the existing classification can be extended into a probabilistic description of subtype progression. We first constructed a longitudinal dataset from the PPMI database by recomputing the subtype classification directly from the underlying MDS-UPDRS items. The resulting dataset contains 1,080 patients and 5,774 visits, of which 4,441 have a valid subtype classification, with follow-up extending to 14 years.

We then examined how patients moved between the three subtypes. The Indeterminate category proved highly unstable: only 19.2% of visits classified as Indeterminate were followed by another Indeterminate visit. We also found that 35.2% of observed subtype changes reversed at the following visit, indicating substantial variability in individual observations. At the same time the movement between subtypes was not symmetric — 253 changes from TD to PIGD against 141 in the opposite direction — indicating a genuine underlying tendency towards PIGD alongside that variability.

These observations motivated a probabilistic approach. A statistical test rejected the assumption that a patient's current recorded subtype contains all the information needed to predict the next one. We therefore used a Hidden Markov Model, which distinguishes a patient's underlying motor state from the subtype recorded at an individual examination. It outperformed the simpler models in all ten validation splits and estimated that the underlying states persist substantially longer than the observed labels suggest.

We also implemented a hierarchical Bayesian model of the continuous TD/PIGD ratio. It estimated an average decline of 0.093 per year (95% credible interval 0.068 to 0.116), giving an interpretable measure of progression, although its predictive accuracy was below that of the Hidden Markov Model.

Overall the results support viewing motor subtype progression as a gradual underlying process observed through an imperfect clinical measurement.

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1 Introduction

1.1 Background

Parkinson's disease is a progressive neurological disorder whose motor symptoms vary considerably between patients. Two broad symptom groups are commonly distinguished. Some patients present predominantly with *tremor*; others present predominantly with difficulties of *gait, balance and postural stability*. The distinction matters clinically because the two presentations are widely believed to differ in their rate of progression and in the risk of later complications.

To make this distinction concrete, patients are assigned to one of three motor subtypes:

- **Tremor Dominant (TD)** — tremor symptoms predominate;
- **PIGD** — gait and postural symptoms predominate;
- **Indeterminate** — neither group clearly predominates.

The assignment is made from the MDS-UPDRS, the standard clinical examination for Parkinson's disease. Eleven of its items measure tremor and five measure gait and postural difficulty. The classification compares the average severity of the two groups.

1.2 The existing classification rule

The comparison is expressed as a single ratio:

$$R = \frac{\text{average score across the 11 tremor items}}{\text{average score across the 5 gait/postural items}}$$

A large value of R means tremor dominates; a small value means gait and postural difficulty dominate. Two fixed thresholds convert the ratio into the three-way label, following the definition of Stebbins et al. (2013):

Ratio	Subtype	Interpretation
$R \geq 1.15$	TD	tremor symptoms predominate
$0.90 < R < 1.15$	Indeterminate	neither group clearly predominates
$R \leq 0.90$	PIGD	gait and postural symptoms predominate

Table 1: The TD/Indeterminate/PIGD classification rule. The Indeterminate band is deliberately placed around $R = 1$: a single hard cut-off would move patients between subtypes on the basis of very small differences, so a buffer zone is used instead.

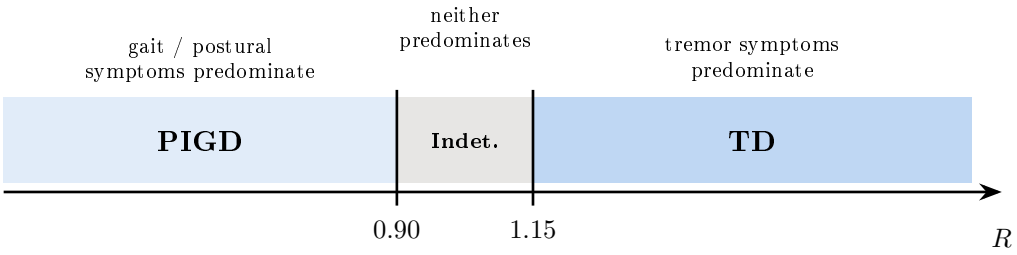


Figure 1: The classification rule as a single axis. **What to notice:** the Indeterminate band occupies only 0.25 units of a scale on which observed values range from 0 to over 11. Section 3.4 shows that this narrowness has consequences.

1.3 Problem statement

The rule above describes a patient at a *single* visit. It answers the question:

"What subtype does this patient appear to have today?"

Parkinson's disease, however, progresses over time, and a patient's recorded subtype can differ from one visit to the next. This raises a question the existing rule cannot answer:

"How does the subtype change over time, and how confident can we be that an observed change represents genuine progression?"

An observed change from TD to PIGD may reflect real disease progression. It may equally reflect ordinary variability in a clinical examination — a slightly different rating on one item, on one day, by one examiner. Distinguishing these two possibilities is the central problem this project addresses.

1.4 Project objective

The objective of this project is to develop a probabilistic understanding of motor subtype progression using longitudinal PPMI data.

This breaks into three concrete aims:

1. Build a reliable longitudinal dataset in which each visit carries a trustworthy subtype label.
2. Establish what the data actually shows about how those labels change over time.
3. Develop models that distinguish genuine progression from variability in the observed labels.

1.5 Research questions

1. How frequently does the recorded motor subtype change between visits, and in which directions?
2. Is the Indeterminate category a stable subtype or a transitional region?
3. Can the observed sequence of subtypes be described by a model that uses only the current label, or is a richer description required?
4. Once measurement variability is accounted for, how stable is a patient's underlying motor state?
5. Does the continuous TD/PIGD ratio show a systematic population-level trend, and at what rate?

1.6 How the project developed

Each stage of this project was motivated by what the previous stage revealed. The report follows that sequence rather than the order in which the software was written.

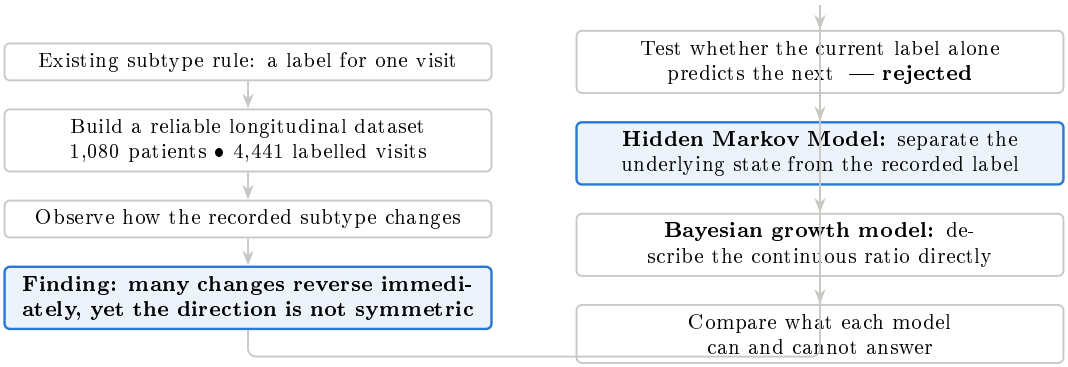


Figure 2: The logic of the project. **What to notice:** the two shaded boxes are the points at which a finding in the data forced a change of method. Neither model was chosen in advance.

2 Data and Motor Subtype Classification

2.1 The PPMI dataset

The Parkinson's Progression Markers Initiative (PPMI) is a large international observational study that follows Parkinson's disease patients over many years, recording a standardised clinical assessment at each visit. It is well suited to this project because the same examination is repeated on the same patients over long periods, which is exactly what is needed to observe how a subtype changes.

We used three components: the Curated Data Cut, which provides the visit structure and patient characteristics; the item-level MDS-UPDRS Part II and Part III files, which provide the sixteen individual scores from which the subtype is computed; and an archived data release, used only to check our work.

2.2 Why the classification was recomputed

PPMI publishes a subtype variable of its own, so a natural question is why we did not simply use it. There were two reasons. First, the published variable in the Curated Data Cut is effectively two-class and does not preserve the Indeterminate category, which — as Section 3.2 shows — turns out to be central to the findings. Second, it does not publish the underlying tremor and gait subscores, and therefore not the continuous ratio, which the model in Chapter 5 requires.

We therefore recomputed the classification from the individual item scores. This also made it possible to verify the result against PPMI's own labels, which is reported in Section 2.5.

2.3 Constructing the longitudinal dataset

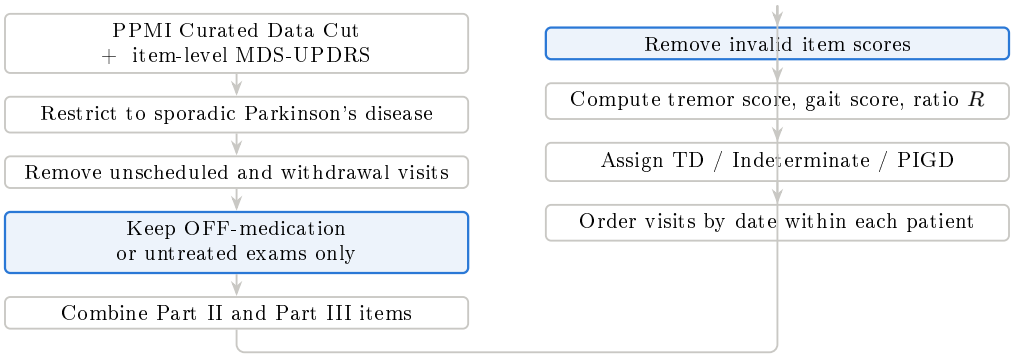


Figure 3: Construction of the longitudinal dataset. **What to notice:** two of these steps — the medication filter and the removal of invalid scores — change the resulting labels substantially and are explained in Section 2.4.

Two aspects of the construction are worth stating explicitly.

Only sporadic Parkinson's disease was included. PPMI also follows patients carrying specific genetic mutations. Those forms of the disease are believed to progress differently, so including them would mix genetic effects into the progression estimates. Restricting to sporadic disease keeps the population homogeneous.

Time was measured from actual visit dates. PPMI labels visits by nominal year, but patients do not attend on schedule. All intervals in this report are computed from recorded visit dates, because the gap between two visits turns out to matter (Section 3.6).

2.4 Data cleaning and two quality issues

Two issues in the raw data affect the resulting classification enough to be worth describing.

Invalid score values. The valid range for an MDS-UPDRS item is 0 to 4. However, PPMI records the value 101 to mean *"Unable to Rate"* — for instance when a patient cannot safely be tested for postural stability — and it is stored in the same numeric field as a real score. Treated as a score, a single such value among the five gait items raises their average to roughly 20, which drives the ratio close to zero and misclassifies a clearly tremor-dominant visit as PIGD. These values were therefore excluded before any score was computed. This is not a rare edge case: the postural-stability item alone carries 4,302 of them.

Medication state. Parkinson's medication suppresses tremor considerably more than it improves gait and balance. An examination performed while a patient is medicated therefore lowers the numerator of the ratio while leaving the denominator largely intact, pushing the patient towards a PIGD classification for reasons that have nothing to do with disease progression. We therefore used only OFF-medication or untreated examinations, which is also what the original definition of the classification specifies. Note that this filters *examinations*, not patients: a medicated patient still contributes through their OFF assessment.

2.5 Validation of the classification

Because the classification was recomputed rather than taken from the database, it had to be checked. We compared our labels against PPMI's own on the visits where both exist.

Reference	Comparison	Agreement
PPMI Curated Data Cut	two-class (TD vs. PIGD)	100.00%
Archived PPMI data release	three-class (incl. Indeterminate)	99.87%

Table 2: Agreement between our recomputed labels and PPMI's own.

Two comparisons are reported because they test different things. The curated variable is two-class, so agreement with it confirms that the tremor and gait subscores and the main threshold are computed correctly, but says nothing about the Indeterminate band. The archived release is the only source in the dataset carrying a three-class label, so it is the only available check on the Indeterminate boundary. Agreement of 99.87% there — the remaining disagreements being visits whose ratio falls essentially exactly on a threshold — establishes that our labels reproduce PPMI's across all three categories.

2.6 The final dataset

Quantity	Value
Patients	1,080
Visits	5,774
Visits with a valid subtype label	4,441 (76.9%)
Consecutive labelled visit pairs (transitions)	3,361
Patients contributing at least one transition	934
Median interval between visits	1.0 year
Median labelled visits per patient	3 (maximum 14)
Maximum follow-up	14.0 years
Subtype distribution, all labelled visits	TD 65.1% • Indet. 9.9% • PIGD 25.0%
Subtype distribution at the first visit	TD 71.7% • Indet. 10.1% • PIGD 18.1%

Table 3: The dataset used throughout this report.

The comparison between the last two rows is already informative: the proportion of PIGD visits is higher across all visits (25.0%) than at the first visit (18.1%), which is a first indication of net movement towards PIGD over time.

3 Observed Motor Subtype Progression

This chapter reports what the data shows before any model is introduced. The findings here are the ones that determined which models were subsequently used.

3.1 Visit-to-visit transitions

For each of the 3,361 consecutive pairs of labelled visits we recorded the subtype at both ends. Expressed as proportions of each starting state:

Current visit	→ TD	→ Indeterminate	→ PIGD
TD	**80.8%**	8.0%	11.2%
Indeterminate	44.0%	**19.2%**	36.8%
PIGD	18.4%	11.4%	**70.2%**

Table 4: Observed transition probabilities. Rows sum to 100%.

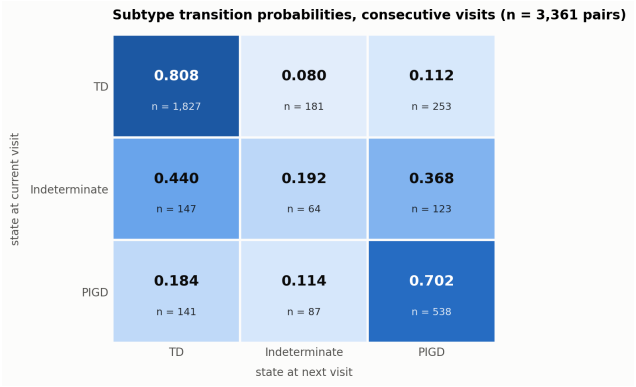


Figure 4: The same information as a heatmap. **What to notice:** the diagonal is dark for TD and PIGD but light for Indeterminate, and the off-diagonal cells in the Indeterminate row are the largest in the table.

3.2 Indeterminate behaves as a transitional region

Finding 1. Indeterminate is not a stable subtype. Only **19.2%** of visits classified as Indeterminate are followed by another Indeterminate visit. The remaining 80.8% move to TD (44.0%) or PIGD (36.8%).

By contrast, TD persists in 80.8% of cases and PIGD in 70.2%. Indeterminate is therefore behaving less like a third subtype and more like a boundary region that patients pass through. This has a direct consequence for modelling: any model that treats the three categories as three comparable states will misrepresent the middle one.

3.3 PIGD is not a one-way destination

A common expectation is that patients progress from TD towards PIGD and remain there. The data does not support the second half of that expectation.

Finding 2. PIGD is not absorbing. **29.8%** of PIGD visits are followed by a non-PIGD classification, and **18.4%** are followed directly by TD.

This should be interpreted carefully. It does not mean that Parkinson's disease reverses. The more plausible reading is that the *recorded* subtype is affected by variability in the examination,

so that some patients recorded as PIGD at one visit were not in fact predominantly PIGD. That interpretation is tested directly in the next section.

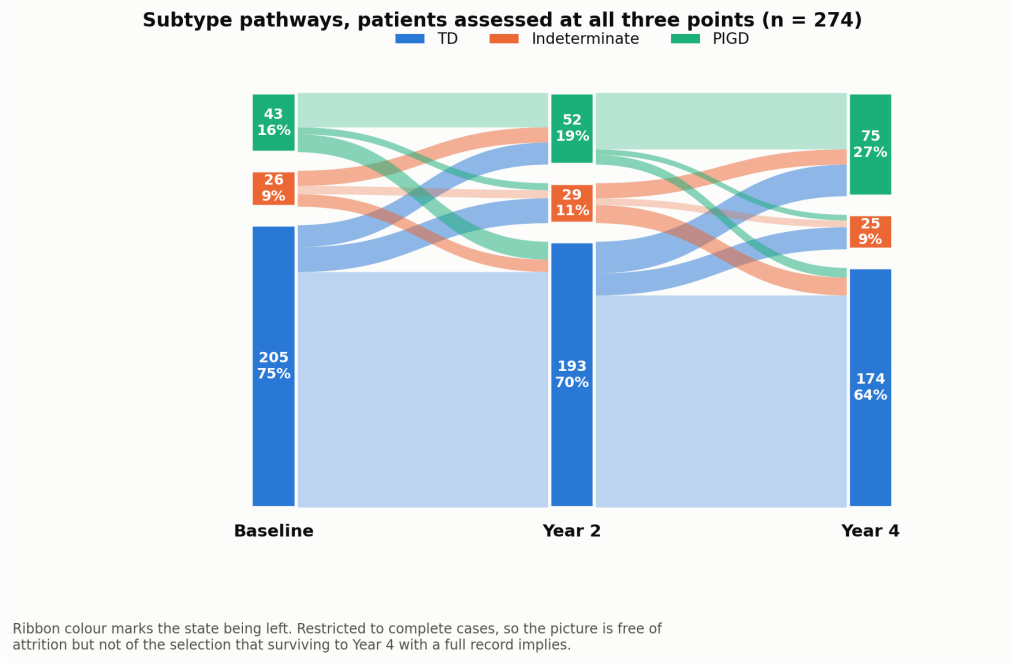


Figure 5: Subtype pathways followed by individual patients over the whole follow-up period, with repeated states collapsed so that only changes are shown. **What to notice:** very few patients follow a single clean transition. Most pathways that reach PIGD pass through Indeterminate, and many return.

3.4 Many observed changes reverse immediately

If a change from one subtype to another reflects genuine progression, we would not expect it to be undone at the very next visit. We therefore took every observed change and asked what happened next.

Finding 3. 35.2% of observed subtype changes reverse at the following visit — the patient returns to the subtype they held before the change.

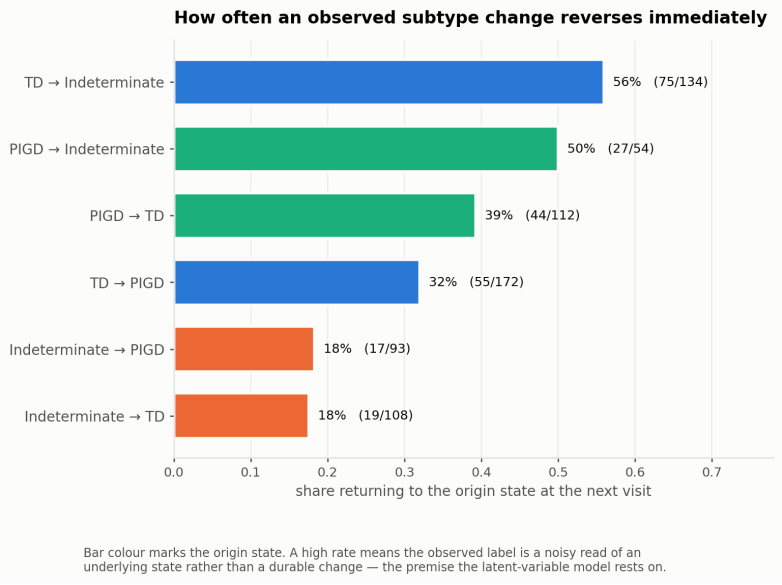


Figure 6: Proportion of changes reversed at the next visit, by type of change. **What to notice:** the rate is not uniform.

The breakdown by direction is more informative than the overall figure:

Observed change	Number	Reversed next visit
TD → Indeterminate	134	**56.0%**
PIGD → Indeterminate	54	**50.0%**
PIGD → TD	112	39.3%
TD → PIGD	172	32.0%
Indeterminate → PIGD	93	18.3%
Indeterminate → TD	108	**17.6%**

Table 5: Reversal rate by type of change. Only changes followed by a further visit can be assessed, giving 673 assessable changes.

Movements *into* Indeterminate reverse about half the time; movements *out of* Indeterminate reverse less than a fifth of the time. This is what one would expect if Indeterminate is a narrow band that patients momentarily appear to occupy because of ordinary measurement variation, rather than a state they genuinely enter and leave. It reinforces Finding 1 from an independent direction.

3.5 But the changes are not merely random

Reversal alone might suggest that observed changes carry no information. The direction of movement shows otherwise.

Finding 4. Movement is directionally biased towards PIGD. Over the whole dataset there were **253** changes from TD to PIGD against **141** from PIGD to TD ($p = 1.8 \times 10^{-8}$).

If observed changes were purely random measurement fluctuations, movement in one direction would be balanced by movement in the other. It is not. Taken together, Findings 3 and 4 give the central conclusion of this chapter:

The data is consistent with **two effects operating simultaneously**: a genuine, gradual, directional progression towards PIGD, *observed through a measurement that carries substantial variability.* Any adequate model must represent both.

3.6 The interval between visits matters

Patients do not attend at regular intervals. We therefore recomputed the transition table separately for short and long gaps between visits.

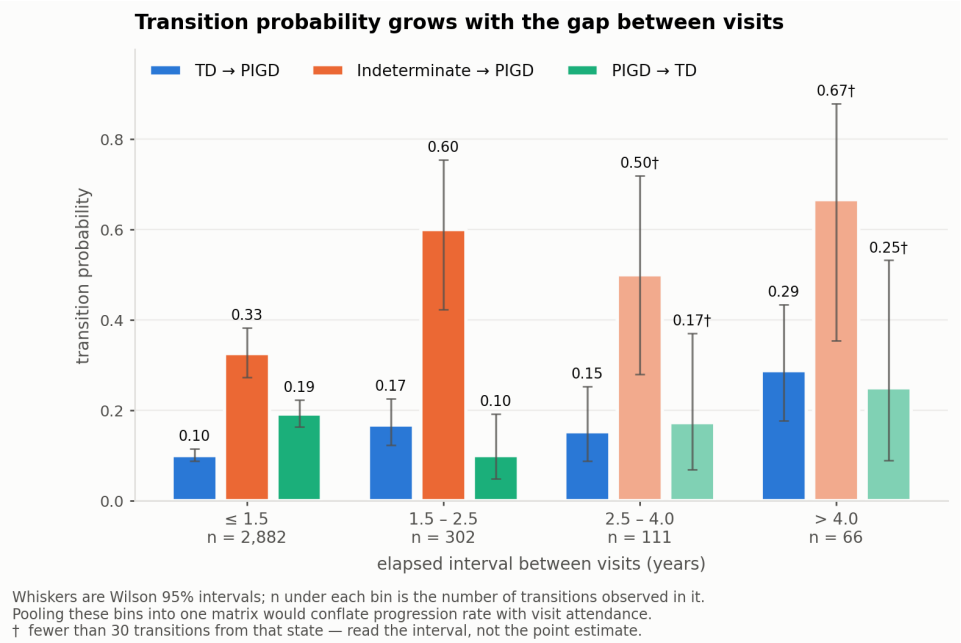


Figure 7: Transition probabilities stratified by the elapsed interval between the two visits. **What to notice:** the probability of leaving the current subtype increases as the gap grows.

A patient reassessed after four years has had far more time in which to change than one reassessed after six months. Treating every visit-to-visit transition as though it spanned the same amount of time is therefore inappropriate, and any model must take the elapsed interval as an input rather than ignoring it.

3.7 Time to PIGD conversion

A natural clinical question is how long it takes for a patient to convert to PIGD. Answering it requires care, because patients have very unequal follow-up: some contributed two visits and some fourteen. Simply reporting the percentage who ever converted would treat those patients as equivalent. We therefore used survival analysis, which accounts properly for patients whose follow-up ends before any conversion is seen.

Definition of conversion	At risk	Converted	5-year incidence
Any single PIGD visit	885	263	35.3%
PIGD sustained over two visits	885	127	15.3%

Table 6: Estimated cumulative incidence of PIGD conversion under two definitions.

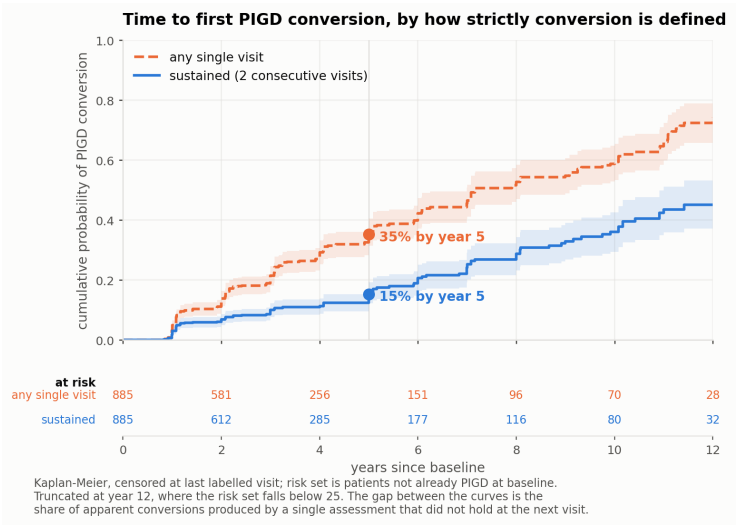


Figure 8: Estimated proportion of patients yet to convert to PIGD, under both definitions. **What to notice:** the two curves separate widely, and the confidence bands broaden at the right-hand end where few patients remain under observation.

The gap between the two rows is itself a finding. Requiring that a PIGD classification persist across two consecutive visits — rather than accepting a single one — more than halves the estimated five-year conversion rate, from 35.3% to 15.3%. Much of the apparent conversion under the looser definition consists of single PIGD visits that are not repeated, which is exactly the pattern described in Section 3.4. Reported conversion rates for this subtype are therefore highly sensitive to a definitional choice that is not always made explicit.

3.8 A limitation with a known direction

Important limitation: missing data is not random

A visit can only be assigned a subtype if all sixteen required MDS-UPDRS items were recorded. This was not the case for **23.1%** of visits in the cohort.

Crucially, the missingness is not random. Items go unrecorded most often when a patient cannot complete the assessment — postural stability being the clearest example — and this occurs more frequently in patients with more advanced disease. Because more advanced patients are also more likely to be PIGD, the visits excluded from this analysis are disproportionately those that would have been classified as PIGD.

Consequence: the progression rates reported throughout this report should be interpreted as **lower bounds**. The true rate of movement towards PIGD is likely to be somewhat higher than estimated here.

4 Probabilistic Modelling of Progression

4.1 The modelling problem

Chapter 3 established that a subtype change may reflect genuine progression or examination variability, and that both are present in the data. The modelling task is therefore:

Given a patient's sequence of recorded subtypes and the times at which they were recorded, describe how their motor state is progressing.

We approached this by testing progressively richer descriptions, beginning with the simplest one that could plausibly work.

4.2 Why a simple model is not sufficient

The simplest description of a sequence of states assumes that the current state contains all the information needed to predict the next one — that once you know a patient is TD today, knowing what they were two years ago tells you nothing more. This is the *Markov assumption*, and it underlies the standard approach to modelling categorical progression.

We tested it directly, using every sequence of three consecutive labelled visits and asking whether the visit before last changes the prediction.

The assumption is rejected, and decisively: $\chi^2 = 156.6$, $p = 7.9 \times 10^{-33}$.

The size of the effect matters as much as its significance. Consider only patients whose current visit is recorded as TD:

Previous visit	→ TD	→ Indeterminate	→ PIGD
TD	84.9%	6.5%	**8.6%**
Indeterminate	52.8%	17.6%	29.6%
PIGD	48.2%	12.5%	**39.3%**

Table 7: Where patients currently recorded as TD go next, according to where they came from.

A patient recorded as TD today has an 8.6% chance of being recorded PIGD next visit if they were also TD last time — but a 39.3% chance if they were PIGD last time. The current label plainly does not contain everything relevant.

This is not merely a negative result. A process in which history predicts the future in exactly this way is the characteristic signature of a system in which the *observed* label is an imperfect reading of some underlying state. The failed test therefore indicates which kind of model is needed.

4.3 Separating the underlying state from the recorded label

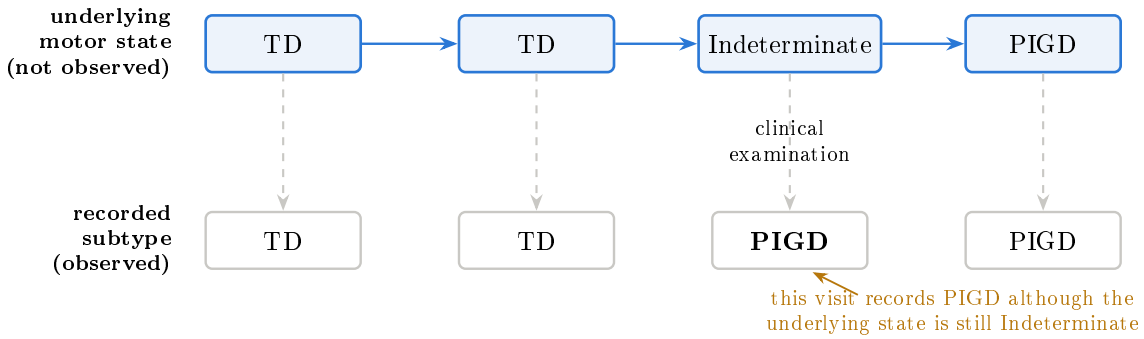


Figure 9: The idea behind the Hidden Markov Model. **What to notice:** the top row progresses smoothly and in one direction; the bottom row — which is all we actually see — appears to jump. The model's task is to infer the top row from the bottom row.

The model we used makes the distinction in Figure 9 explicit. It supposes that each patient has an underlying motor state which changes gradually over time, and that each clinical examination produces a recorded subtype which usually — but not always — matches it. The model estimates two things at once from the data: how quickly underlying states change, and how reliably each one is recorded.

This is implemented as a **Hidden Markov Model** (HMM). "Hidden" refers to the underlying state, which is never observed directly. Because the intervals between visits are irregular (Section 3.6), the underlying state is modelled as changing in continuous time, so the elapsed gap between two visits enters the model directly.

For comparison we also fitted three simpler models: a Markov chain applied to the observed labels; the same chain made interval-aware; and a null model that ignores the current state entirely and predicts the overall subtype distribution. The null model exists to establish a floor — any model that cannot beat it is not using the information available.

4.4 Model validation

All models were compared on patients whose data was withheld during fitting. The split was made *by patient*, not by visit: every visit belonging to a given patient was assigned entirely to either the training or the test set. Splitting by visit would allow the same patient to appear on both sides and would make the models look better than they are.

Model	Accuracy	Brier score
Hidden Markov Model	**76.9%**	**0.346**
Markov chain on observed labels	76.7%	0.363
Interval-aware Markov chain	76.5%	0.367
Null model (no use of state)	62.5%	0.519

Table 8: Performance on 1,002 held-out transitions from patients not used in fitting. Brier score measures the quality of the predicted probabilities, and lower is better; accuracy alone rewards only the single most likely prediction.

Accuracy separates the models only slightly, because all of them get the easy cases right — a TD patient usually remains TD. The Brier score separates them more clearly, because it evaluates the whole predicted probability distribution rather than only the most likely outcome. This is the more relevant measure for a model intended to express uncertainty.

The Hidden Markov Model outperformed the plain Markov chain in **all ten** independent splits, by an average margin of 0.028 in held-out log-likelihood per transition (standard deviation 0.009). The improvement is small but entirely consistent.

4.5 What the model tells us

The model's predictive advantage is modest. Its interpretative value is larger, because the quantities it estimates answer questions the raw transition table cannot.

Finding 5: progression is mostly sequential. The estimated rates of change between underlying states are:

$$\text{TD} \to \text{Indeterminate}: \ 0.094 \text{ per year} \qquad \text{TD} \to \text{PIGD}: \ 0.006 \text{ per year}$$

Movement from TD directly to PIGD is estimated to be roughly sixteen times rarer than movement from TD into Indeterminate. The model therefore describes progression as passing *through* the intermediate region rather than jumping across it — consistent with the pathway structure seen in Section 3.2, and reached here by an entirely separate route.

Finding 6: the recorded label is much less reliable for Indeterminate. The model estimates, for each underlying state, how often an examination records it correctly:

Underlying state	Recorded correctly
TD	94.1%
PIGD	84.2%
Indeterminate	**50.0%**

Table 9: Estimated probability that an examination records the underlying state correctly.

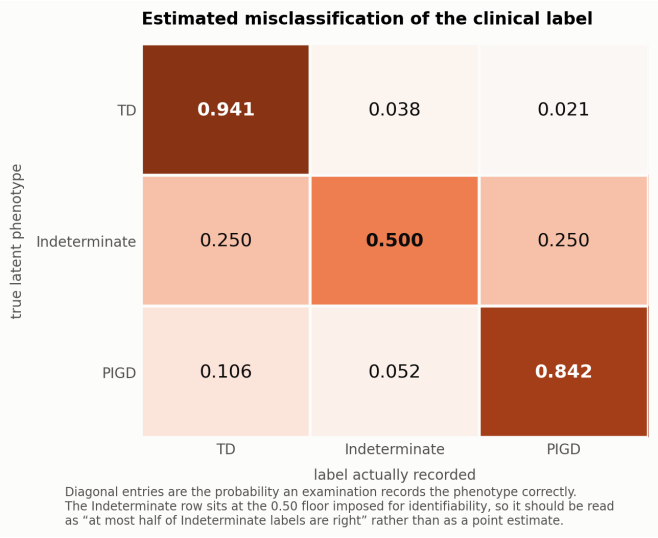


Figure 10: Estimated relationship between underlying state and recorded label. **What to notice:** the TD and PIGD rows are concentrated on the diagonal; the Indeterminate row is spread across all three columns.

TD and PIGD are recorded reliably. Indeterminate is recorded correctly only about half the time. This is a third independent line of evidence pointing at the same conclusion as Findings 1 and 3, and there is a simple explanation for it: the Indeterminate band is only 0.25 units wide (Figure 1), which is small relative to the variation observed in the ratio between consecutive visits of the same patient.

Finding 7: underlying states are more stable than the labels suggest. Once measurement variability is accounted for, the model estimates how long a patient remains in an underlying state before genuinely moving on:

State	From observed labels	Estimated underlying state
TD	3.4 years	**10.1 years**
Indeterminate	0.4 years	**2.1 years**
PIGD	2.3 years	**16.5 years**

Table 10: Estimated persistence of a motor state, before and after accounting for variability in the examination.

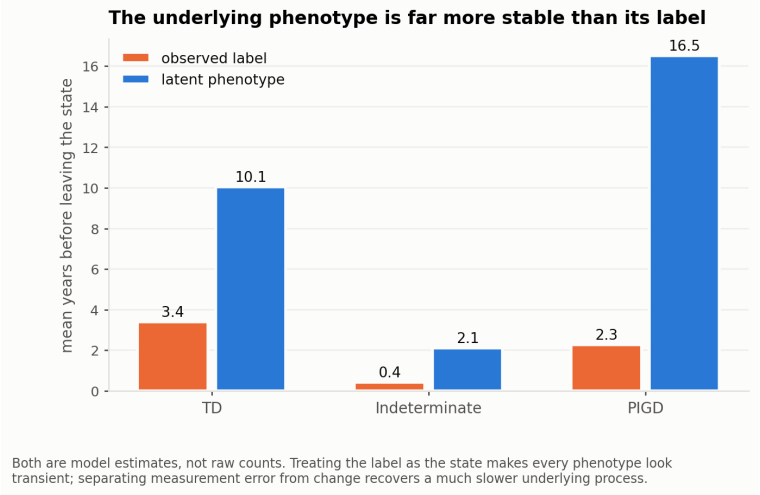


Figure 11: Persistence of each motor state as read from the observed labels against the model's estimate for the underlying state. **What to notice:** the gap is large for every state, and largest for PIGD.

The underlying motor state appears to be roughly three to seven times more persistent than the recorded subtype suggests. On this reading, most of the apparent instability seen in Chapter 3 is variability in the examination rather than genuine change in the patient.

This is the most substantial claim in the report, and it should be read as the model's interpretation of the data rather than as a directly measured quantity. Its plausibility rests on the fact that three independent analyses — the persistence of Indeterminate, the pattern of immediate reversals, and the estimated reliability of the recording — all point the same way.

5 A Bayesian Model of the TD/PIGD Ratio

5.1 Why a second model

The model in Chapter 4 works with the three discrete subtypes. But the classification is derived from a continuous quantity — the ratio R — and converting it into three categories discards information. A patient whose ratio falls from 4.0 to 2.0 has changed considerably, yet remains TD throughout and contributes nothing to a transition-based analysis.

We therefore asked a different question:

"Does the ratio itself show a systematic trend over time, and how fast?"

This is a question about *rate*, which the discrete model cannot answer at all. It was also the modelling approach suggested in the original project proposal.

5.2 The idea of the model

The model describes each patient's ratio as following its own straight line over time:

$$R_{it} = \underbrace{\text{patient's starting level}}_{\text{differs between patients}} + \underbrace{\text{patient's rate of change}}_{\text{differs between patients}} \times t + \underbrace{\text{variation}}_{\text{differs between visits}}$$

Three ideas are combined here. Each patient has their own starting level and their own rate. Those individual values are not free to be anything: the model assumes they are drawn from a population distribution, so that a patient with few visits is described mostly by the population average and a patient with many visits is allowed to depart from it. And each individual measurement is allowed to deviate from the patient's own line.

This structure is called a **hierarchical Bayesian growth model**. The "hierarchical" part is the assumption that individual patients come from a common population; the "Bayesian" part means the result is not a single estimate but a distribution expressing how certain we are about each quantity.

Four quantities are estimated: the population starting level, the population rate of change, how much patients differ from one another, and how much individual visits vary around a patient's own line.

5.3 The population progression rate

Estimated population progression rate

$\beta_1 = -0.0927$ per year 95% credible interval $[-0.1159, -0.0682]$

The TD/PIGD ratio declines by approximately 0.093 per year on average. The estimated probability that the population rate is negative — that is, that the population as a whole is drifting towards PIGD — is essentially 100%.

A "credible interval" can be read in the natural way: given the model and the data, there is a 95% probability that the population rate lies between 0.068 and 0.116 per year.

To put the rate in context: from the estimated population starting level of 2.29, an average trajectory takes approximately **12.3 years** to fall to the TD threshold of 1.15. Progression at the population level is therefore a slow drift measured in decades, not a rapid conversion.

5.4 Variation between patients

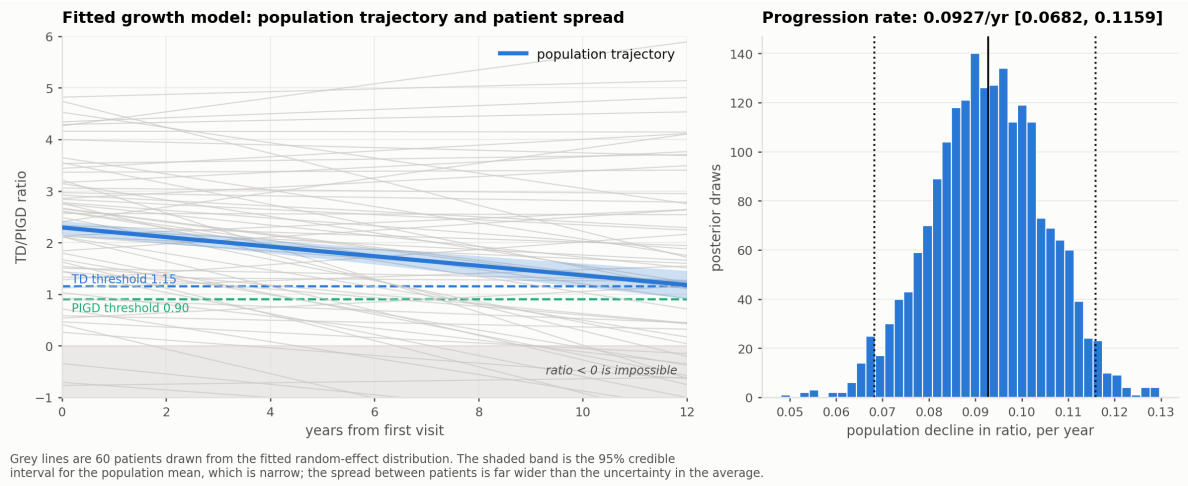


Figure 12: Left: the estimated population trajectory with individual variation. Right: the estimated progression rate.

Figure 12 requires explanation, because the two panels show different kinds of quantity and are easily confused.

The blue line and band (left panel). The blue line is the estimated *average* trajectory across the population. The shaded band around it expresses how uncertain we are about that average. It is narrow, because the estimate rests on thousands of observations.

The grey lines (left panel). The grey lines are *example individual patients* generated from the estimated variation between patients. They are not real patients, and they are not intended to pass through any observed measurement. They illustrate the range of individual trajectories that the fitted model considers plausible.

Why the grey lines are straight. Because the model describes each patient by a straight line, every trajectory drawn from it is straight by construction. Real patients' measurements do not lie on straight lines — they vary considerably from visit to visit — and that variation is represented in the model by the third term of the equation, which is simply not drawn in this figure. The figure shows modelled trends, not observations.

The right panel. This shows the estimated population rate of change as a distribution rather than a single number. The solid line marks the central estimate and the dotted lines the bounds of the 95% credible interval. The entire distribution lies well away from zero, which is the visual form of the statement that the population is progressing.

What the comparison shows. The essential point of the left panel is the contrast between the narrow blue band and the wide grey fan. We know the population *average* precisely, and individual patients *differ from it a great deal.* The estimated spread of individual rates is wide enough that a minority of patients are drifting in the opposite direction, towards TD. These are two different

things, and only the first is reduced by collecting more data; the second is genuine variation between people.

5.5 Two important limitations

Implementing this model surfaced two limitations that are worth stating plainly, because they bear on how the result should be used.

Two sources of variation cannot be separated. The model as originally specified distinguished variation arising from the disease process itself from variation arising from measurement. With one measurement per visit, these two cannot be estimated separately: they enter the observed ratio in exactly the same way, so the data can determine their combined size but not how it divides between them. We therefore report the combined variation only. (In statistical terms, the two components are not *identifiable*.) This matters because a statement about the underlying ratio ideally refers to the disease process alone.

The assumed distribution does not match the data. The model assumes each measurement varies symmetrically around the patient's line, following a normal distribution. The ratio does not behave this way. It cannot be negative; it is strongly skewed towards high values; and it is exactly zero at 7.0% of visits, in patients with no measurable tremor. A consequence is that the model places **12.2%** of its five-year predictions below zero, which is impossible for a ratio of non-negative scores. The growth model should therefore be read as a description of the average trend, not as a predictive model for individual patients.

5.6 Comparison with the Hidden Markov Model

Both models were evaluated on the same held-out transitions — restricted to the 825 that both are able to score, since the growth model cannot be applied to visits where the ratio is undefined.

	Accuracy	**Brier score**
Hidden Markov Model	**76.1%**	**0.357**
Bayesian growth model	73.7%	0.379

Table 11: Performance on 825 shared held-out transitions. (These figures are not directly comparable with the 76.9% in Chapter 4, which was measured on the full set of 1,002 held-out transitions.)

There is a measurable reason for the difference, and it is not a matter of one model being better designed. The visit-to-visit variation in the ratio is several times larger than the width of the Indeterminate band it must predict membership of. Even a perfectly estimated trajectory therefore leaves the next observation able to fall in any of the three categories. The discrete model avoids this problem entirely by never attempting to predict a continuous value.

The two models should not be read as competing attempts at the same task. The Hidden Markov Model is better suited to describing the sequence of motor states and to separating the underlying state from the recorded label. The Bayesian growth model is the only one of the two that provides an interpretable rate of progression with a stated uncertainty.

	Hidden Markov Model	Bayesian growth model
Question answered	Which state is the patient in?	How fast is the ratio changing?
Works with	Three discrete subtypes	The continuous ratio
Handles noisy labels	Yes, explicitly	Only as combined variation
Gives a rate	No	Yes, with an interval
Uses all visits	Yes (4,441)	No (3,716)
Held-out accuracy	76.1%	73.7%

Table 12: What each model is suited to.

6 Overall Findings

6.1 Summary of findings

Finding	Evidence
The recorded subtype changes frequently	19.2% of TD, 80.8% of Indeterminate and 29.8% of PIGD visits are followed by a different classification
Indeterminate is a transitional region, not a stable subtype	Only 19.2% of Indeterminate visits are followed by another
Many observed changes reverse immediately	35.2% overall, rising to 50–56% for changes *into* the Indeterminate band
The direction of change is nonetheless systematic	253 changes TD $\rightarrow$ PIGD against 141 in the opposite direction ($p = 1.8 \times 10^{-8}$)
The current label does not fully describe the patient's state	Markov assumption rejected, $\chi^2 = 156.6$, $p = 7.9 \times 10^{-33}$; prior state moves the probability of PIGD next visit from 8.6% to 39.3%
Separating underlying state from recorded label describes the data better	The Hidden Markov Model wins all ten independent validation splits
Underlying states are more persistent than the labels suggest	An estimated 10.1 years for TD and 16.5 for PIGD, against 3.4 and 2.3 as read directly from the records
The continuous ratio declines at the population level	0.093 per year (95% credible interval 0.068–0.116), reaching the TD threshold after approximately 12.3 years

6.2 Where the evidence converges

The strongest conclusions are those supported by more than one independent analysis.

Conclusion	Independent lines of evidence
Indeterminate is a boundary region, not a subtype	Persistence of only 19.2%; reversal rate of 50–56% for moves into it; estimated recording reliability of 50.0%; and progression modelled as passing through it rather than around it
The recorded label carries substantial noise	35.2% immediate reversal; failure of the Markov assumption; the gap between observed and estimated underlying persistence; and the halving of the estimated conversion rate when two consecutive visits are required
Progression is genuine and directed towards PIGD	Asymmetry of 253 against 141; the population ratio declining at 0.093 per year with essentially 100% posterior probability; and PIGD prevalence rising from 18.1% at the first visit to 25.0% across all visits

Table 13: Conclusions supported by multiple independent analyses.

6.3 What remains uncertain

- **How much of the observed instability is measurement variation.** The estimate that underlying states are three to seven times more persistent than the records suggest is a model-based inference. Its direction is well supported; its magnitude depends on the model.
- **Whether progression rates are constant over time.** All models here assume a constant rate. There is evidence in the data that the rate of movement towards PIGD is higher later in follow-up than earlier, which the current models do not represent.
- **Individual rates of progression.** The population rate is estimated precisely; an individual patient's rate is not. With a median of three visits per patient, an individual estimate is dominated by the population average rather than by that patient's own data.
- **Whether the findings hold outside PPMI.** No external cohort was used.

7 Limitations

1. **Incomplete and non-random labelling.** 23.1% of cohort visits could not be assigned a subtype. Items go unrecorded most often in patients who cannot complete the assessment, who tend to have more advanced disease and to be PIGD, so the progression estimates here are lower bounds.
2. **An assumption in the Hidden Markov Model.** The model requires a minimum reliability for the recorded label; without it the fitting procedure returns a degenerate result. This minimum is an input rather than an estimate, although the conclusions do not change materially across a range of values for it.
3. **The growth model's distributional assumption.** The assumed normal distribution does not match a quantity that is non-negative, right-skewed and exactly zero at 7.0% of visits.
4. **Unseparated sources of variation.** In the growth model, process variation and measurement variation cannot be estimated separately from the available data.
5. **Study population and external validity.** Only sporadic Parkinson's disease patients from PPMI were included, and all validation used held-out patients from within that same cohort. No independent study was used.

6. **Constant rates and irregular timing.** All models assume progression rates do not change over time, and treat visit timing as uninformative. Neither assumption is exactly correct.

8 Conclusion and Future Work

8.1 Conclusion

This project began from a limitation of the existing motor subtype classification: it describes a patient's subtype at a given visit but says nothing about how that subtype changes over time.

By constructing a longitudinal dataset from PPMI — recomputing the classification from the underlying examination items and validating it at 100.00% and 99.87% against PPMI's own labels — and then examining the observed transitions, we found that subtype changes contain substantial variability. The Indeterminate category behaves more like a transitional region than a stable state, and more than a third of observed changes reverse at the following visit. At the same time the movement is clearly asymmetric, so the variability coexists with genuine directional progression.

These findings motivated a probabilistic approach. A statistical test showed that a patient's current recorded subtype does not contain all the information relevant to predicting the next one — the signature of an underlying state being observed imperfectly. A Hidden Markov Model built on that idea described the data better than the simpler alternatives in every validation split, and estimated that the underlying motor states persist substantially longer than the recorded labels suggest. It also indicated that progression occurs mostly through the Indeterminate region rather than directly from TD to PIGD.

A hierarchical Bayesian growth model addressed the complementary question of rate, estimating an average decline in the TD/PIGD ratio of 0.093 per year with a 95% credible interval of 0.068 to 0.116. This provides an interpretable measure of progression, although its distributional assumptions limit its use for individual prediction.

Taken together, the results support viewing motor subtype progression as a **gradual underlying process observed through an imperfect clinical measurement**. A single observed subtype change is weak evidence of progression; a sustained one is considerably stronger.

8.2 Future work

Model the two component scores rather than their ratio. The clearest methodological improvement available. The ratio's denominator is the average of only five items, and in this cohort its typical value is small enough that a one-point change on a single item can move the ratio dramatically. Modelling the tremor and gait scores jointly would remove that instability, avoid the impossible negative predictions, and recover the 725 visits currently excluded because the ratio is undefined — visits that are overwhelmingly TD and therefore not missing at random.

Allow progression rates to vary over time. There is evidence that the rate of movement towards PIGD is not constant across follow-up. Relaxing this assumption is a direct extension of the current models.

Predict progression for a newly presenting patient. The models in this report use a patient's history. A clinically more useful question is what can be said at a patient's *first* visit, when no history exists. Preliminary work in this direction has been carried out and will be reported separately.

Validate on an independent cohort. All results rest on a single study. Confirming them on an independent longitudinal Parkinson's cohort is necessary before the findings can be regarded as general.

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A Subtype Calculation

Tremor score. The mean of eleven MDS-UPDRS items: one from Part II (2.10, tremor) and ten from Part III (3.15a–b postural tremor; 3.16a–b kinetic tremor; 3.17a–e rest tremor amplitude; 3.18 constancy of rest tremor).

Gait/postural score. The mean of five items: two from Part II (2.12 walking and balance; 2.13 freezing) and three from Part III (3.10 gait; 3.11 freezing of gait; 3.12 postural stability).

Classification. With T the tremor score and P the gait/postural score, $R = T/P$ and

$$\text{subtype} = \begin{cases} \text{TD} & R \geq 1.15, \text{ or } P = 0 \text{ and } T > 0 \\ \text{Indeterminate} & 0.90 < R < 1.15, \text{ or } T = P = 0 \\ \text{PIGD} & R \leq 0.90 \end{cases}$$

A subscore is undefined if any of its constituent items is missing, matching PPMI's own rule. The ratio is undefined when $P = 0$; the classification is still well defined in that case, which is why the discrete models can use 4,441 visits while the growth model can use only 3,716.

B Data Processing Rules

1. Retain patients in the sporadic Parkinson's disease subgroup.
2. Exclude unscheduled and premature-withdrawal visits.
3. Exclude visits beyond 15 years of follow-up.
4. Retain MDS-UPDRS Part III examinations recorded as OFF medication, or belonging to untreated patients.
5. Where a visit carries more than one examination, retain the record with the fewest missing items.
6. Treat any item value outside 0–4 as missing. This removes the sentinel value 101 ("Unable to Rate") and any future sentinel.

7. Compute elapsed time from recorded visit dates. PPMI publishes month and year only, so intervals carry a resolution of approximately one month.
8. Require at least two labelled visits for a patient to contribute a transition.

C Full Transition Counts

From	→ TD	→ Indet.	→ PIGD	Total
TD	1,827	181	253	2,261
Indeterminate	147	64	123	334
PIGD	141	87	538	766
Total	2,115	332	914	3,361

Table 14: Observed transition counts.

From	To	Probability	95% lower	95% upper
TD	TD	0.808	0.791	0.824
TD	Indeterminate	0.080	0.070	0.092
TD	PIGD	0.112	0.100	0.126
Indeterminate	TD	0.440	0.388	0.494
Indeterminate	Indeterminate	0.192	0.153	0.237
Indeterminate	PIGD	0.368	0.318	0.421
PIGD	TD	0.184	0.158	0.213
PIGD	Indeterminate	0.114	0.093	0.138
PIGD	PIGD	0.702	0.669	0.734

Table 15: Transition probabilities with Wilson 95% intervals. Wilson intervals are used rather than the standard normal approximation because the latter can produce bounds outside $[0, 1]$ when a probability is close to zero or one or the count is small — which occurs in five cells of the interval-stratified table.

D Model Specifications

Hidden Markov Model. A latent state $S(t) \in \{\text{TD}, \text{Indeterminate}, \text{PIGD}\}$ evolves as a continuous-time Markov chain with generator Q, so that the probability of moving between states over an elapsed time t is $P(t) = e^{Qt}$. Each observation is drawn from the latent state through a matrix E, where E_{ij} is the probability that true state i is recorded as label j. The likelihood of a patient's full sequence marginalises over every possible latent path. The model has 14 free parameters: six transition rates, six recording probabilities and two parameters for the initial state distribution.

Estimated rates (per year)	→ TD	→ Indet.	→ PIGD
TD	—	0.0937	0.0057
Indeterminate	0.1156	—	0.3557
PIGD	0.0000	0.0605	—

Table 16: Estimated rates of change between underlying states.

Hierarchical Bayesian growth model. For patient i at time t,

$$R_{it} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})\, t + \varepsilon_{it}, \qquad b_{0i} \sim \mathcal{N}(0, \tau_0^2),\ b_{1i} \sim \mathcal{N}(0, \tau_1^2),\ \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$$

Estimated by Gibbs sampling with weakly informative priors on all parameters.

Parameter		Mean	2.5%	97.5%
β_0	population starting level	2.2920	2.1577	2.4272
β_1	population rate, per year	−0.0927	−0.1159	−0.0682
σ	visit-to-visit variation	1.1380	1.0993	1.1778
τ_0	spread of starting levels	1.3408	1.2466	1.4421
τ_1	spread of rates, per year	0.1033	0.0780	0.1315

Table 17: Posterior summaries. Fitted on 2,426 observations from 573 training patients. The estimates were stable across five different prior specifications.

The comparison of $\sigma = 1.138$ against the Indeterminate band width of 0.25 is the quantitative reason the growth model predicts subtype less well than the discrete model: visit-to-visit variation is roughly 4.6 times the width of the band whose membership must be predicted.

E Validation Results

Split	Test pairs	HMM	Interval Markov	Markov	Null
0	997	−0.6522	−0.6691	−0.6679	−0.8446
1	1,051	−0.6604	−0.6919	−0.6863	−0.8730
2	1,045	−0.6891	−0.7211	−0.7198	−0.8981
3	994	−0.6960	−0.7196	−0.7108	−0.8810
4	967	−0.6585	−0.6929	−0.6870	−0.8585
5	982	−0.6917	−0.7314	−0.7303	−0.8897
6	1,047	−0.6803	−0.7259	−0.7217	−0.8968
7	993	−0.6529	−0.6997	−0.6901	−0.8610
8	959	−0.7322	−0.7604	−0.7516	−0.8955
9	1,020	−0.6672	−0.7012	−0.6986	−0.8664
Splits won by the HMM					**10 of 10**
Mean margin over the Markov chain					+0.0284 **(sd** 0.0094**)**

Table 18: Held-out log-likelihood per transition on ten independent patient-level splits. Higher (less negative) is better. The ordering of the models is identical in every split.

F Reproducibility

All results in this report are reproducible from the project repository. Every analytic choice — item lists, thresholds, cohort rules, medication state, random seeds — is recorded in a single configuration file rather than in code, so that each can be audited or varied without modification to the analysis. The analysis is covered by an automated test suite.

Stage	Produces
Dataset construction	The labelled longitudinal panel and the validation against PPMI's own labels (Chapter 2)
Transition analysis	Transition tables, reversal rates, pathways and conversion curves (Chapter 3)
Progression models	The Markov assumption test, the fitted models and all held-out comparisons (Chapter 4)
Bayesian growth model	Posterior summaries, trajectories and the comparison against the Hidden Markov Model (Chapter 5)

Table 19: Analysis stages and the results each produces.

Data used in the preparation of this report were obtained from the Parkinson's Progression Markers Initiative (PPMI) database. For up-to-date information on the study, visit `www.ppmi-info.org`.